

Lakshmi Bharathy Kumar

AI Engineer with strong Data Science foundations

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Authorized to work in the U.S. without employer-based visa sponsorship

Summary

AI Engineer with an M.S. in Applied Data Intelligence (San Jose State University, May 2026), 2 years of analytics experience, and strong Data Science foundations. Builds production-grade LLM pipelines, agentic systems, and causal inference models – from data engineering through deployment and observability. Comfortable owning the full stack in a small team.

Technical Skills

Languages: Python, SQL

Agentic AI & LLMs: LangGraph, LangChain, MCP (Model Context Protocol), RAG, Prompt Engineering, Pydantic, Instructor, Groq/OpenAI APIs

Evaluation & Observability: Ragas, LangSmith, MLflow, Streamlit

ML, Statistics & Causal Inference: scikit-learn, XGBoost, SHAP, Statsmodels, A/B Testing, Uplift Modeling, Bayesian Modeling, Feature Engineering

Data Engineering & Cloud: Airflow, Snowflake, dbt, AWS (SAA-C03 in progress – Aug 2026), Docker, FastAPI, Git, CI/CD

Experience

Digital Marketing Specialist

Jun 2021 – Dec 2022

RepuNEXT

Chennai, India

- Analyzed campaign and website performance using Google Analytics and SEMrush to identify funnel drop-off points; updated SEO targeting based on findings, growing organic traffic 25% and improving ROI 15%.
- Built automated weekly reporting workflows for campaign metrics and audience segmentation; advised clients on social media content strategy in recurring review meetings.

Projects

AI Customer Support Triage Agent | *Python, LangGraph, FAISS, Groq, SQLite, Docker*

- Production B2B SaaS ticket triage system with an 8-node LangGraph graph and tiered SLA enforcement (Gold/Diamond/Platinum). Tier-routing and SLA deadline logic are deterministic code, fully decoupled from LLM calls – business-critical rules stay auditable regardless of model behavior.
- Hierarchical parent-child RAG over 30 articles; faithfulness-gated auto-resolution (≥ 0.85) with genuine LangGraph interrupt/checkpoint for HITL review. Measured: faithfulness 0.81, relevance 0.94, \$0.00139/ticket (~\$1.32/1,000).

Enterprise LLM Observability & Evaluation Engine | *Ragas, LangSmith, MLflow, FastAPI, Python*

- Automated Ragas evaluation pipelines tracking Faithfulness (avg 0.81), Answer Relevance, and Context Recall across releases; per-node telemetry for token usage and latency; cut API costs 18% via prompt deduplication and caching. Deployed as a FastAPI service.

Autonomous Data Validation Pipeline | *Python, Pydantic, Instructor, SQLite*

- Vendor files arrived unformatted and inconsistent at scale; built an ingestion and validation pipeline enforcing strict schema and type-safety on LLM outputs via Pydantic + Instructor, with a self-healing retry loop (3 cycles max) that raised data consistency from 50% to 94%.

Marketing Mix & Incrementality Modeling for Ad Spend Efficiency | *Python, Google Meridian, scikit-learn, XGBoost*

- Built a Bayesian MMM using Google Meridian across 5 channels on 156 weeks of geo-level data; VIF-informed prior tightness per channel (Channel3 VIF=20.99) handled collinearity where data alone could not isolate individual contributions; media mix contributed 21.8% of total conversions.
- Recommended \$4.8M reallocation (+39.7% to highest-ROI channel, -15.7% from lowest-mROI), projecting +0.5% lift. Uplift models (T-learner, X-learner) on 1M+ users: naive targeting wasted ~22% of spend; uplift-based targeting captured 14% more incremental conversions at equal budget.

Customer Lifetime Value Segmentation & Churn Risk | *Python, XGBoost, SHAP, scikit-learn, Power BI*

- The top 18% of 96,000+ customers drove 61% of revenue – surfaced via RFM feature engineering and K-Means clustering. XGBoost churn classifier (ROC-AUC 0.84) with SHAP flagging recency and purchase-gap as leading signals; CLV regression showed At-Risk customers had 3.2x higher projected value than Lost.

Education

San Jose State University

San Jose, CA

M.S., Applied Data Intelligence

May 2026

- Thesis – Benchmarked 5 ML classifiers and 2 fine-tuned CNNs on eye-tracking data; best model reached 0.94 cross-user accuracy (LOUO), 12pp above published baseline. Manuscript in preparation.

Saranathan College of Engineering

Trichy, India

B.E., Electronics and Communication Engineering

May 2014